

NH₃ Converter Unit

CU-2

CODE: I0041296000A

⚠ WARNING	
	ELECTRIC SHOCK
	MAINTAIN GROUND TO AVOID
	ELECTRIC SHOCK.
	Do not ground this unit to a gas pipe or to an indirect ground, such as another piece of equipment.

WARNING

Before being shipped from the factory this unit was carefully adjusted. Do not touch internal controls or switches unless readjustment becomes necessary.

WARRANTY

HORIBA, Ltd., as manufacturer, warrants each instrument to be free from defects in material and workmanship under normal use and service for a period of one (1) year after delivery to the original purchaser, the manufacturer's obligations under this warranty being limited to repairing or replacing, at its option, any part or parts thereof which shall, within one (1) year after delivery of such unit to its original purchaser, be returned by the original purchaser to the manufacturer at its factory or authorized repair center, transportation charges prepaid, and which upon examination by the manufacturer, shall be determined to the manufacturer's satisfaction to be defective. The warranty shall cover all parts and labor necessary to make repairs for the first year delivery. The warranty shall be limited to the original purchaser.

The provisions of this warranty shall not apply to any unit which has been subject to misuse, negligence or accident in installation or operation or which shall have been repaired, altered or serviced in any manner or which shall not have been operated by the original purchaser in a manner specified by the manufacturer so as, in the manufacturer's judgment, to adversely affect its operation. The original purchaser shall, upon request of the manufacturer, furnish manufacturer reasonable evidence that the defect arose from causes placing a liability upon manufacturer. If the warranty shall not apply, the original purchaser shall pay all repair and replacement costs and all costs of manufacturer in returning the part to the original purchaser. This warranty is expressly in lieu of any other warranties, expressed or implied, including any implied warranty of merchantability or fitness of a particular purpose, and/or any other manufacturer by others bear the warranty of their manufacturer.

HORIBA, Ltd., reserves the right to make changes in the design and price of its equipment at any time and to exclude certain components from this warranty without prior notice.

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1 . DESCRIPTION

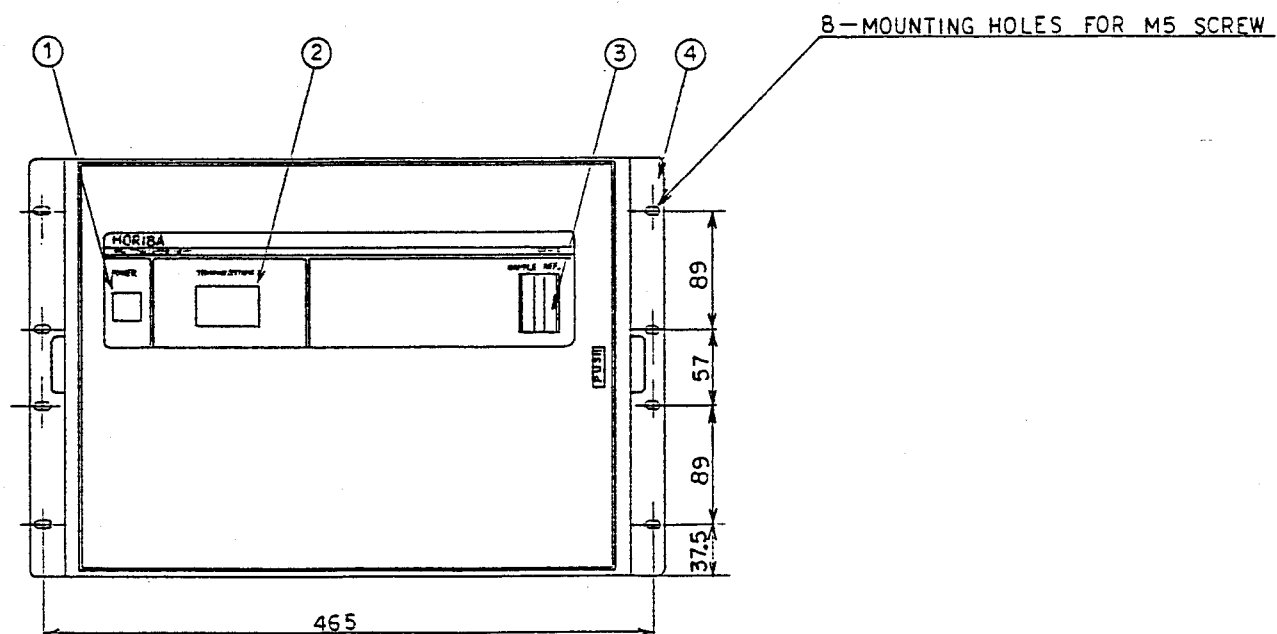
1.1Introduction

CU-2 is preconditioning unit to measure NH_3 in ambient air in combination with NO_x analyzer.

In CU-2 unit, NH_3 is converted into NO_x to measure as NH_3 .

1.2 Descriptions of Components

1.2.1 Front panel



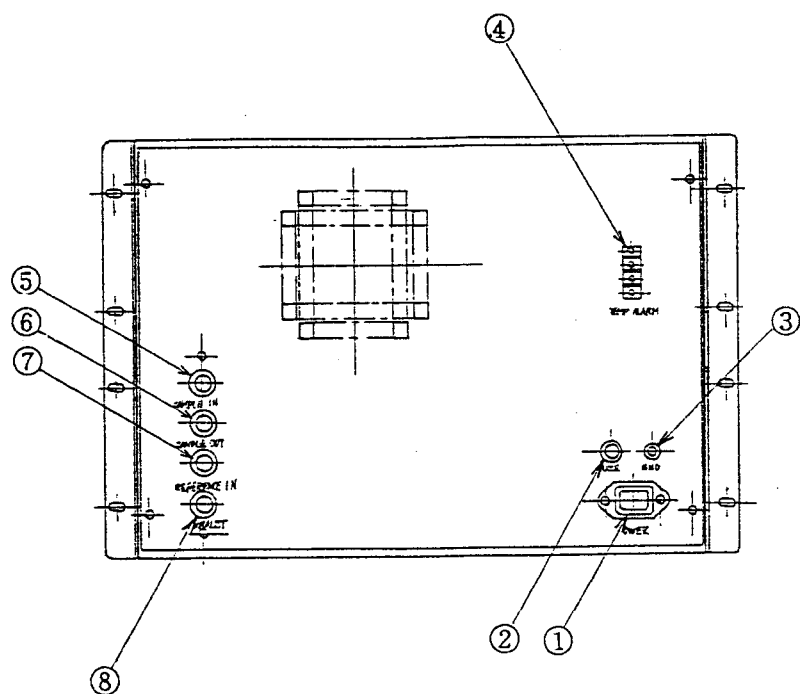
①Power switch

②Display panel

③Flow meter

④Fixing plate(option)

1.2.2 Rear panel



①Power connector

②Fuse

③GND terminal

④Alarm terminal

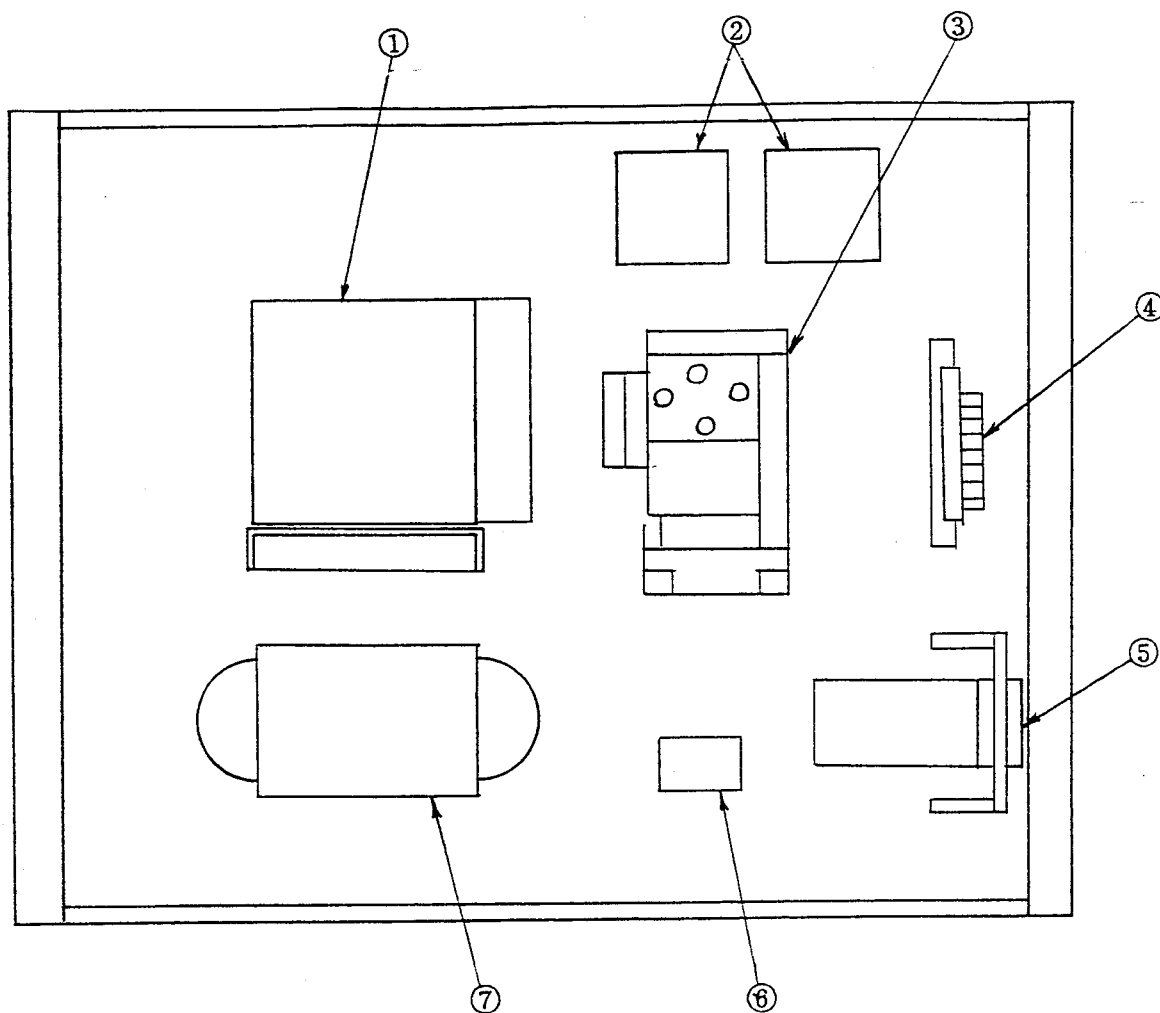
⑤Sample inlet

⑥Sample outlet

⑦Reference outlet

⑧Exhaust

1.2.3 Inside the unit



①Catalyzer

②NO_x converter

③Pump

④Sample gas filter

⑤Thermoregulator

⑥Relay

⑦Transformer

2.SPECIFICATION

Principle:	NH ₃ is converted into NO _x to measure as NH ₃ .
Application:	NH ₃ in ambient air
Flow rate:	Approx. 4.0 L/min
Oxidation efficiency:	90% or more
Interference:	Within ±0.01ppm at NO,NO ₂ 0.5ppm
Output:	Alarm:Temperature
Power:	100,115 or 220V AC approx.80VA(as specified)
Ambient temperature:	0-40 ℃
Dimensions:	430(W)*310(H)*450(D) mm
Mass :	Approx. 30kg

3.PRINCIPLE

Normally, NO_x is contained in sample gas in addition to NH_3 .

Filtered gas flow is divided into sample line and reference line.

In sample line, NH_3 is oxidized to NO_x by catalyst.

This NO_x and NO_x originally contained are converted to NO in NO_2 to NO converter unit.

The sum of NH_3 , NO_2 and NO is measured in sample line.

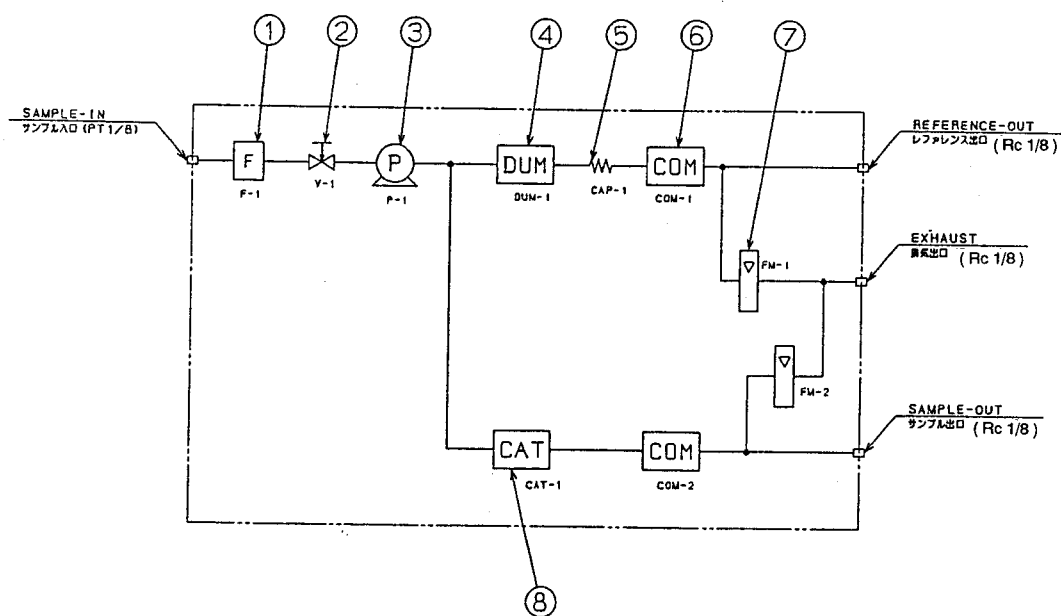
In reference line, sample gas is directly introduced to NO_2 to NO converter.

NO_x (sum of NO and NO_2) only is measured in reference line.

Hence, difference between sample line and reference line corresponds to

NH_3 content in sample gas.

Flow Sheet



① Filter

② Needle valve

③ Pump

④ Dummy

⑤ Capillary

⑥ Converter

⑦ Flow meter

⑧ Catalyzer

4. INSTALLATION

4.1 Location

The instrument should be located in a clean area, free of:

- A) Excessive dust
- B) Mechanical vibrations
- C) Strong electric or electro-magnetic fields
- D) Corrosive gasses, and/or
- E) Moisture exceeding 90%

Choose a location where sudden temperature changes in excess of 5°C do not occur, and where the temperature does not exceed the range of 0 to 40 °C.

Avoid any location where the instrument would be exposed to direct sunlight or radiation from heaters.

CAUTION:

The unit is designed for use in a general purpose area and is not rated for use in hazardous areas.

4.2 External Dimensions

No.	Part Name	Notes
①	POWER SWITCH 電源スイッチ	
②	DISPLAY PANEL 表示パネル	
③	FLOW METER 流量計	
④	FIXING PLATE 固定板	OPTION
⑤	POWER CONNECTOR 電源コネクタ	
⑥	FUSE ヒューズ	
⑦	GROUND TERMINAL アース端子	
⑧	ALARM TERMINAL 警報端子	
⑨	SAMPLE INLET サンプル入口	
⑩	SAMPLE OUTLET サンプル出口	
⑪	REFERENCE INLET リファレンス入口	
⑫	EXHAUST 排気口	

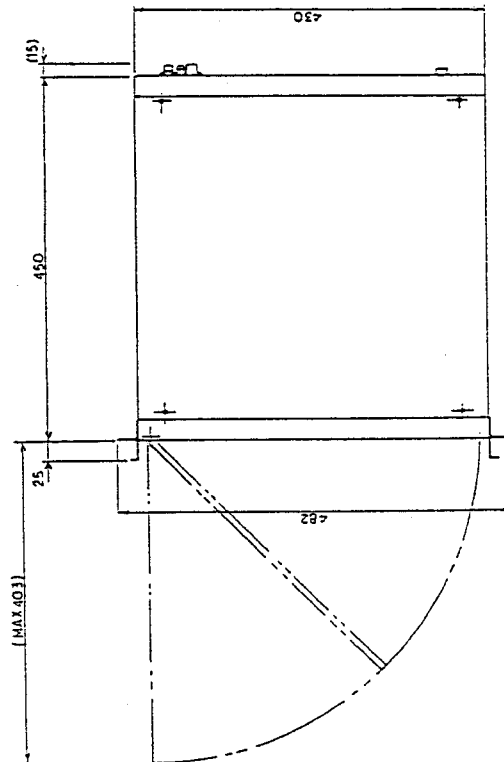
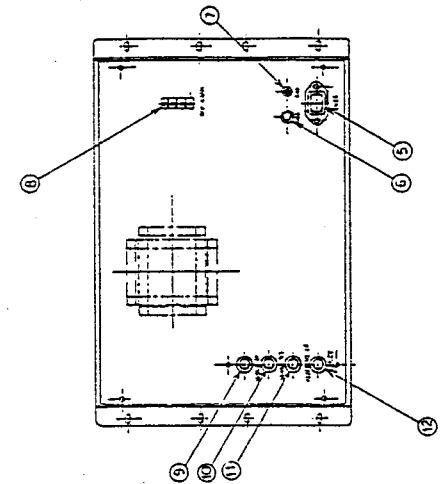
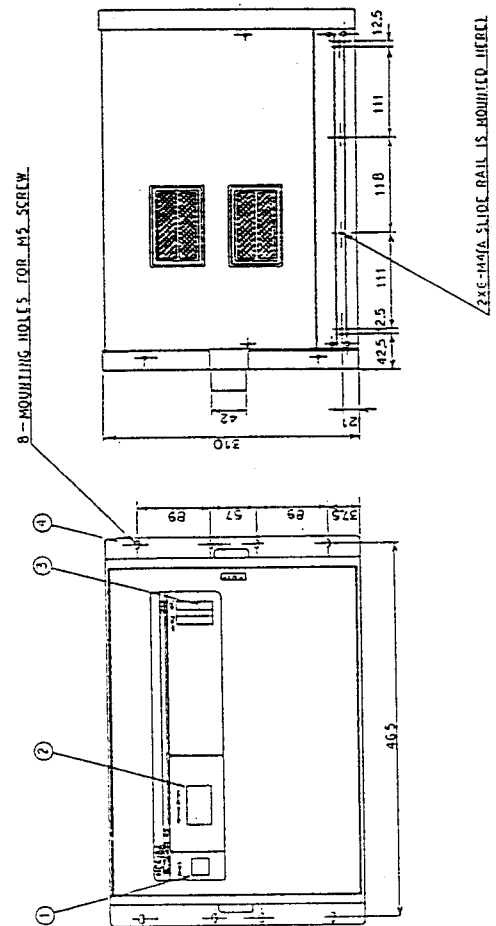


Fig. 4-1 External Dimensions



[UNIT: mm]



4.3 Mounting

The instrument is designed for mounting on:

- A) A standard 19 inch rack,
- B) A free-standing instrument panel, or
- C) A flat table top.

The unit requires 310 mm of rack space or a panel cutout as recommended in Fig. 4-2 and 4-3.

For A) or B), four mounting screws suited to the rack or the panel or which the instrument is to be mounted will be required.

Mounting brackets and external slide rails are available as optional extras.

The fixing plate on the both sides of the instrument (for rack or panel mounting) may be removed when installing the instrument on a table top.

The use of supporting bracket(s) underneath the instrument base plate is recommended for rack or panel installations.

Allow adequate space behind and in front of the instrument to give service access.

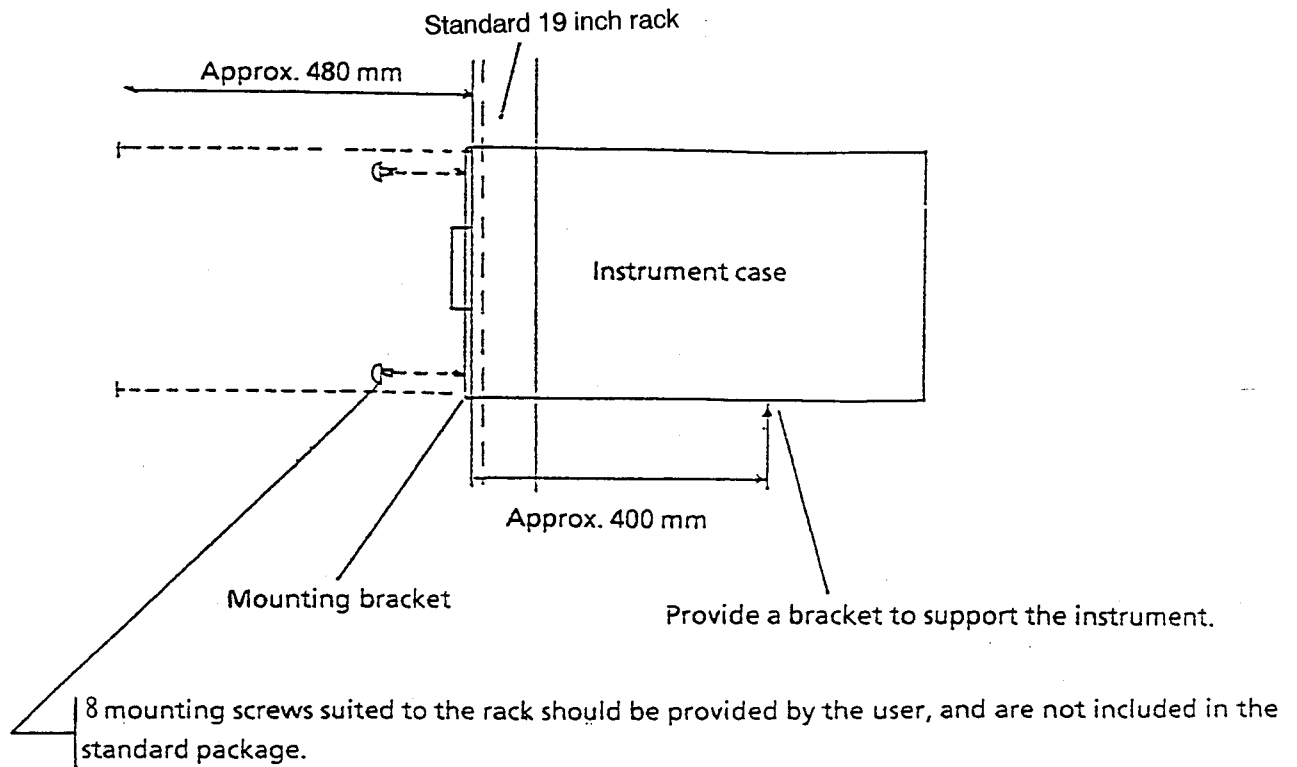
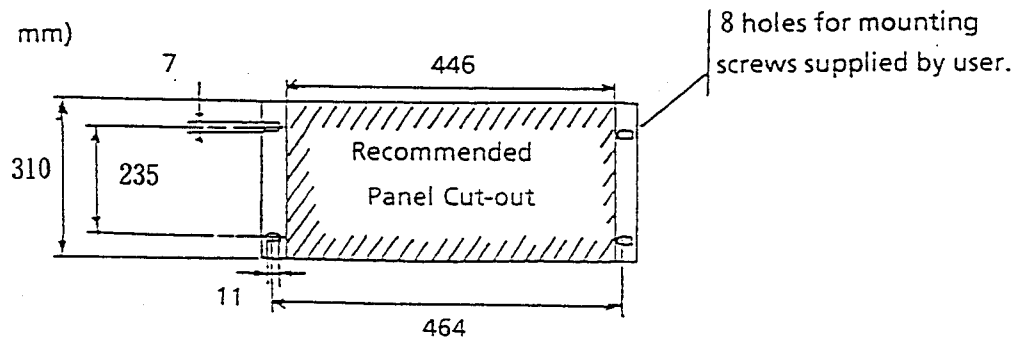


Fig. 4-2 Mounting on Rack (A)

The recommended panel cutout and a screw positions are shown in the following illustration.

(UNIT: mm)



Provide a supporting bracket at rear of the instrument case as shown in Fig. 4-2 Mounting on Rack (A).

Note: Fixing plate and slide rails are available as optional extras.

Fig. 4-3 Mounting on Panel (B)

4.4 Piping

The instrument is shipped with all internal plumbing installed.

Use of fluorocarbon tubing is recommended for all external piping connections

Utilize adequate tube connectors at the connector fittings on the instrument for all external piping connections

* Tube connectors are not included in the standard package.

The threads of the fittings are Rc 1/8. (See Fig. 4-4)

All lines must be leak-free. For leakage inspection do not use soap solution.

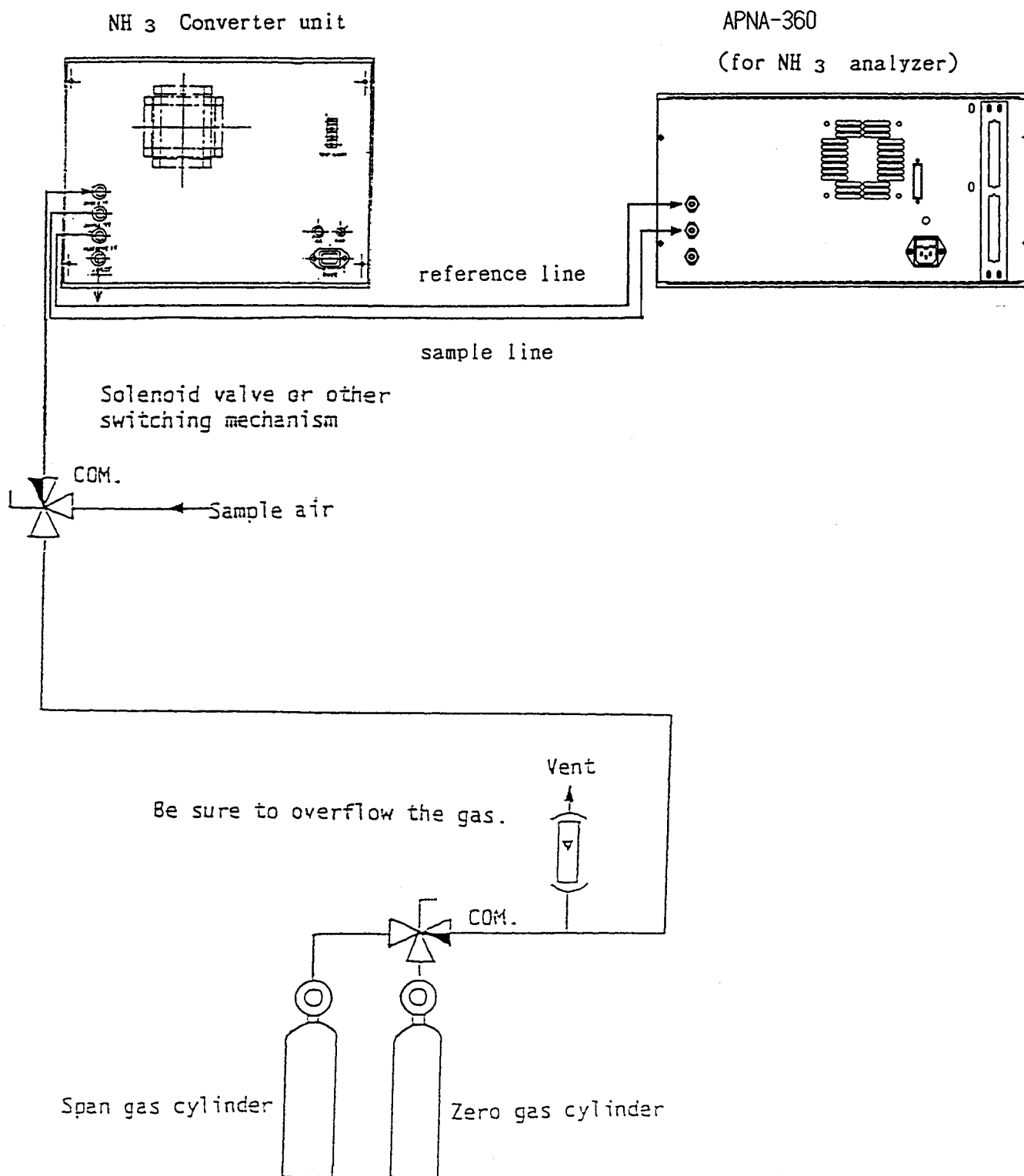


Fig. 4-4 Piping

4.5 Calibration Gases

4.5.1 Zero gas

Provide a Zero gas cylinder.

Use air for the zero gas.

4.5.2 Span gas

Provide a Span gas cylinder.

Use air-balanced NH_3 for the span gas.

4.6 Sampling Precautions

The sample gas must satisfy following requirements at the sample inlet of the instrument:

A) Impurities in the sample gas

Corrosive gases such as SO_3 , Cl_2 , F_2 , or HCl should not be present, and should be removed from the sample prior to entering the instrument. The presence of these substances in the sample will adversely affect performance and may shorten life of materials in contact with the sample

B) Sample gas temperature

The sample gas must be maintained at a temperature within 0 to 40°C at the sample inlet of the instrument. The dew point should be maintained at least 5°C lower than the ambient temperature to prevent condensation in the instrument or sampling system.

C) Back pressure

The instrument outlet should not have a back pressure of more than ± 0.5 kPa.

Excessive back pressure may cause an erroneous reading or damage to the instrument mechanism.

4.7 Wiring

Power:

Before connecting power to the monitor, verify that the front panel POWER switch is in the "OFF" position and that FUSE holder on the instrument rear panel is furnished with the proper fuse.

The power cord for the instrument ends in a three-prong jack which plugs into a recessed receptacle label POWER.

The monitor should be connected to a "CLEAN" power line having an external breaker capable of providing the electricity as indicated at POWER receptacle.

CAUTION:

Avoid installing the instrument near any equipment which excessively distorts the power waveform or generates surges. These include electric welders, high-frequency induction furnaces and large motors. Also avoid sharing a power line with such equipment.

DO NOT CONNECT the instrument to any power line of non-specified voltage and/or frequency, or the unit may be damaged.

5. OPERATION

5.1 Warming up

Make sure that the installation, piping and wiring have been carried out properly, as described in Chapter 4, before turning the power ON.

Then turn ON the power switch to connect the power.

The TEMP. alarm will normally extinguish in approximately 1 h after supplying the power.

Leave the equipment to warm up for approximately another 40 min after the TEMP. alarms go out.

Note: A longer warming up period is required if the ambient temperature is below 10°C.

5.2 Zero and Span Calibration

1. After warm-up, check to make sure that none of the alarms on the front panel are lit.
2. Pass Zero gas into the sample inlet "SAMPLE IN" on the rear panel. Make sure that the gas overflows from the standard gas generator before passing it into the unit.
3. Pass Span gas into the sample inlet "SAMPLE IN". Make sure that the gas overflows from the standard gas generator before passing it into the unit.

5.3 Alarm Outputs

An alarm signal is output from the terminal board on the rear panel of the CU-2.

ALARM : Temp Alarm

The contact ratings of the alarm signals are all DC 30V, 0.5A.

The CU-2 is equipped with a catalyzer. When there is an abnormality with the temperature of the unit, the Temp alarm goes on.

Set temperature $\pm 30^\circ\text{C}$

When the temperature goes outside this range, the Temp alarm goes ON.

The alarm stays ON for approximately one hour after the power is turned ON.

6. MAINTENANCE

6.1 Zero and Span Calibration (once every 1 - 2 weeks)

Pass Zero or Span gas from the sample inlet "SAMPLE IN" and adjust to the appropriate concentrations with the Zero control or Span control. Refer to 5.2 Zero and Span Calibration.

6.2 Replacing Filter Elements (once every 1-2 weeks)

Replace both the sample gas filter and Reference gas filter elements on a regular basis.

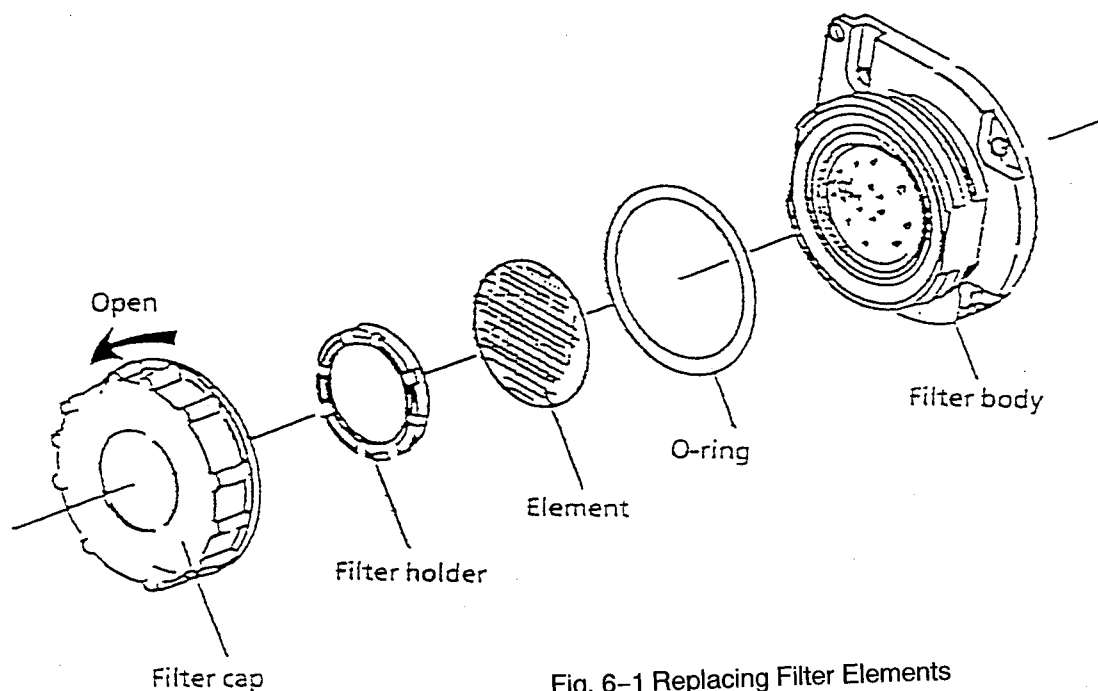


Fig. 6-1 Replacing Filter Elements

Fully tighten the caps after replacing the elements, to prevent leaks.

6.3 Replacing the Pump Diaphragm (every 6 months)

- Shut down the instrument.
- Disconnect tubings and remove the pump from the instrument. Put the pump on a clean table top.
- Remove screws ① from cover plate ② of pump and disassemble the pump, referring to Figure below. Take the parts out from the top. (See Fig. 6.2)
- Replace valve-read ③ after removing screw-ring ④.
- Replace diaphragm ⑤ after removing screw from piston rod ⑥.

- f) Replace diaphragm ⑤ and related parts with new ones that are packaged in a diaphragm kit.
- g) Reassemble pump and replace it in its original position in sampling unit.
- h) Reconnect the tubings.
- i) Replace cover of pump unit.

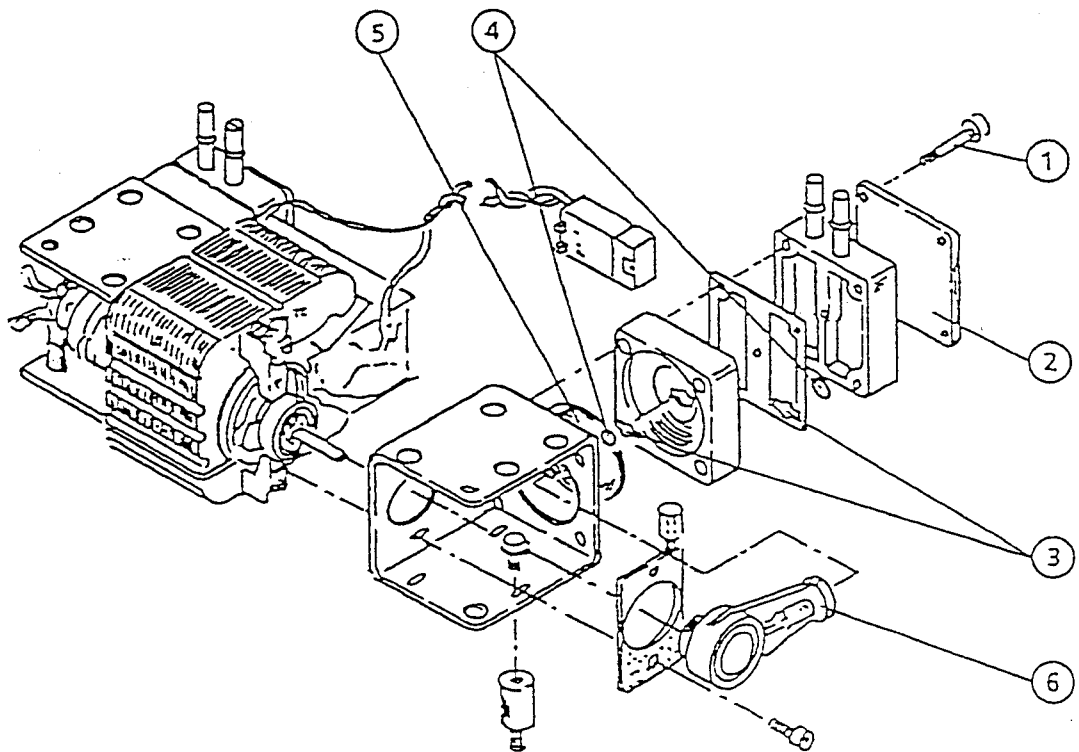


Fig. 6-2 Replacing the Pump Diaphragm

6.4 Replacing the Catalyzer Pipe Unit of the NO_x Converter (once per year)

- * Open the rack cover in order to replace this unit.
- * Use a long screwdriver.

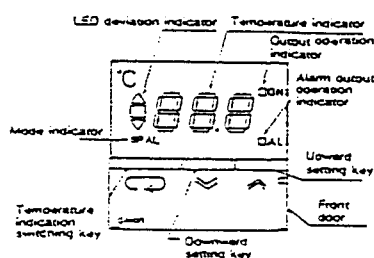
After a long use, the efficiency (NO₂ - NO) of the converter drops. Consequently, the catalyzer pipe unit (S310321 ⑧) should be replaced once every 1 year. (The period of replacement varies according to the condition of the gas.)

6.5 Replacing the Catalyst (once per year)

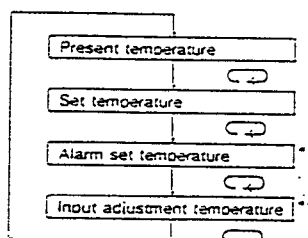
See S210353

6.6 Thermoregulator Adjustment

■ NAMES OF RESPECTIVE PARTS

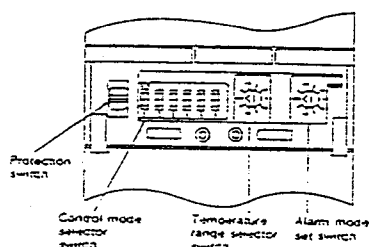


• Temperature indication



- NOTES:**
- Alarm set temperature is available only with type ESCS-□□□.
 - When alarm mode set switch is set to "0" or "9", no alarm temperature will be displayed.
 - Not displayed when the control mode selector switch No.4 is OFF.

• Function selector switch



- NOTES:**
- Downward and Upward setting keys are locked when Protection switch is on "PROTECT" side.
 - Functions are not changed when switching the position of control mode selector switch, temperature range selector switch or alarm mode set switch during energization. Turn off the power before doing.
 - Alarm mode set switch is available only for the type ESCS-□□□.

- Setting the DIP switches
- Be sure to check settings of internal DIP switches before operation. If the setting condition is different, control action will be altered.
- After setting is made, fill in the blank label provided e. g. temperature range for reference.
- When using the product as "F" version apply the "F" unit label provided to "C" unit on the panel.

■ CONTROL MODE SELECTOR SWITCH

		DIP switch No.					
		1	2	3	4	5	6
Control mode	PID	ON					
	ON/OFF	OFF					
Proportional period	2Sec	ON					
	20Sec	OFF					
Control action	Normal			ON			
	Reverse			OFF			
Input adjustment	Possible			ON			
	Impossible			OFF			
JIS/DIN	DIN					ON	
	JIS (= ISO)					OFF	
°C/°F	°F						ON
	°C						OFF
Factory set specification		OFF	OFF	OFF	OFF	OFF	OFF

NOTES

- The optimum proportional band is set and offset is adjusted automatically in the PID action.
- When you desire to promote the input shift function, set the input shift value to "0". (The factory-set condition is "0").

• Thermocouple input type (ESCS-□□KJ)

SW No.	°F	°C
0	*	0-200
1	*	0-300
2	0-400	0-400
3	0-500	0-500
4	0-500	0-500
5	0-999	0-999
6	*	0-200
7	*	0-300
8	0-400	0-400
9	0-500	0-500

• Platinum resistance thermometer type (ESCS-□□P)

SW No.	°F	°C
0	*	-50-50
1	*	0-50.0
2	*	-20-80
3	0.0-99.9	0.0-99.9
4	0-200	0-200
5	*	0-300
6	0-400	0-400
7	0-500	*
8	0-500	*
9	*	*

• Thermistor type (ESCS-□□G)

SW No.	°F	°C
0	*	-50-50
1	*	0-100
2	*	50-150
3	*	100-200
4	*	150-300
5	-50-100	*
6	0-300	*
7	100-300	*
8	200-400	*
9	300-500	*

- NOTES:**
- Never set the switch to the positions with *.
 - The switch No. marked with * is factory set specification at shipment.

■ ALARM MODE SET SWITCH

SW No.	Operation
0: No alarm output	OFF
1: Upper- and lower-limit alarms	X X
2: Upper-limit alarm	X
3: Lower-limit alarm	X
4: Upper- and lower-limit range alarm	X X
5: Upper- and lower-limit alarm with standby sequence	X X
6: Upper-limit alarm with standby sequence	X
7: Lower-limit alarm with standby sequence	X
8: Absolute value alarm	X

A: Set temperature, X: Alarm set value.

■ SELF-DIAGNOSTIC FUNCTION

- The display shows the following indications when failure occurs.

Display	Failure	Output state	
		Reverse action	Normal action
---	Under flow	ON	OFF
FFF	Over flow	OFF	ON
FFF or --- (flashing)	Input sensor failure	OFF	OFF
E11	Memory back-up failure	OFF	OFF
E33	A/D converter failure	OFF	OFF

- NOTES:** Input sensor failure display is not available for thermistor input type (ESCS-□□G).

7. PARTS LIST,EXPLODED VIEWS AND OTHER RELATED DRAWINGS

Chapter 7 contains the electrical schematics and parts lists for the CU-2.

Exploded views with referenced parts list

Catalyzer unit S 2 1 0 3 5 3

NOx converter unit..... S 3 1 0 3 2 1

Flow schematic..... V 3 4 4 8 1 6 B (CU-2)

V 3 5 9 2 1 4 B (APNA- 3 6 0 f o r
 N H ₃ a n a l y z e r)

Wiring diagram..... V 3 4 4 9 2 1 C

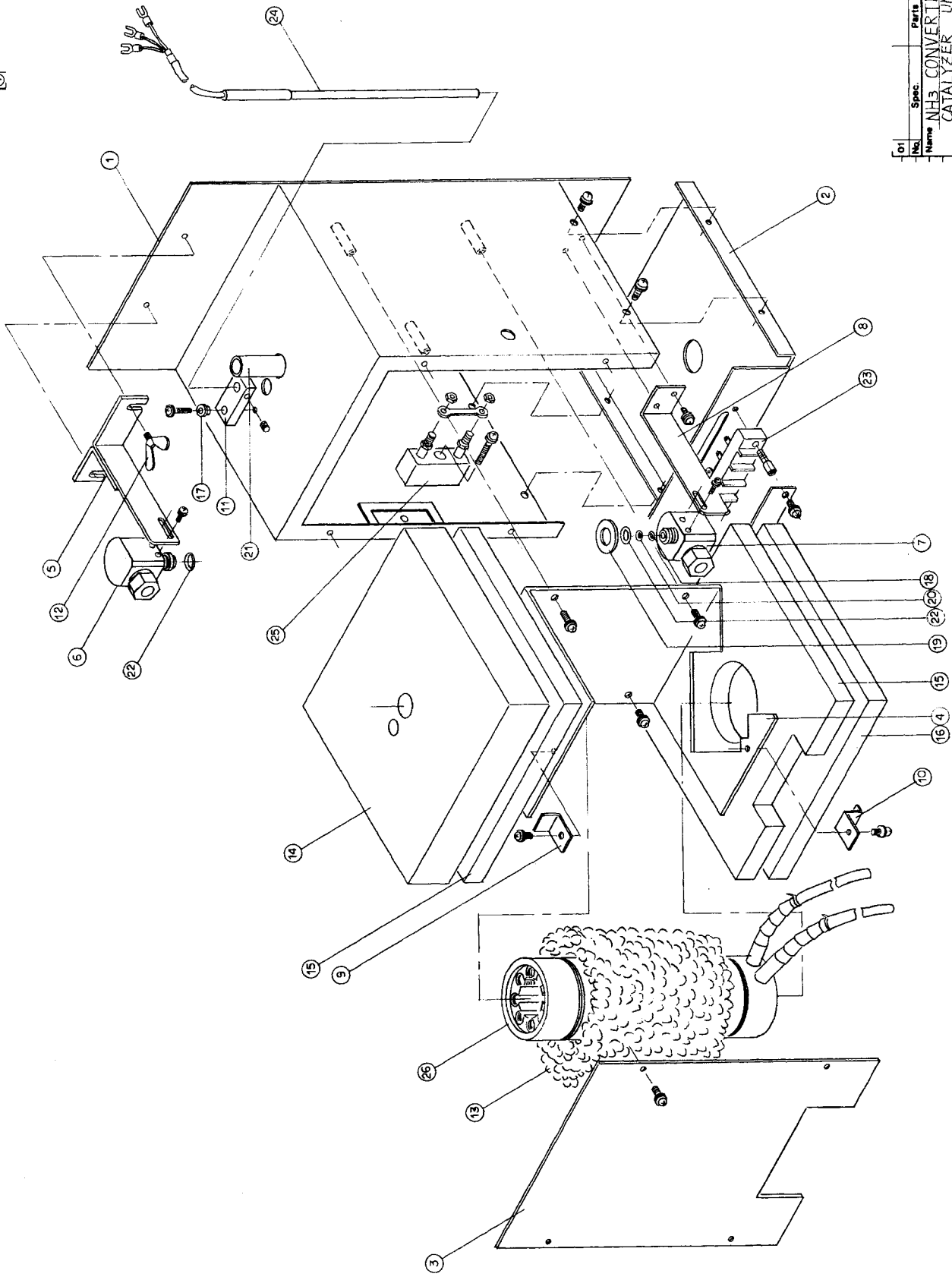
Catalyzer unit S 2 1 0 3 5 3

NO.	Parts Number	Parts Name	Description
1	H347689-01	Reactor case	
2	H347690-01	Cover	
3	H518334-01	Cover	
4	H347691-01	Heater plate	
5	H347692-01	Fixing plate	
6	H518295-01	Fixing joint	
7	H518296-01	Fixing plate	
8	H518297-01	Plate	
9	H518298-01	Plate	
10	H518299-01	Holder	
11	F020165100	Screw	
12	F020341700	Thermal insulator	
13	H518337-01	Thermal insulator 1	
14	H518338-01	Thermal insulator 2	
15	H518338-02	Thermal insulator 3	
16	H487723-01	Spacer	
17	H431850-09	Filter	
18	H514415-03	Packing sheet	
19	F020064400	O-ring	P4(FKM)
20	U 8 0 0 8 4 1 5 0 0	Qualz tube	L=225 OD=15
21	F021412800	O-ring	P8(SI)
22	E010616900	Terminal	
23	E011762400	Thermocouple	
24	E011720900	Terminal	
25	E012038100	Thermal fuse	S13-200V-15A-230℃
26	H325431A01	Electrical furnace	

NO_x Converter unit

S 3 1 0 3 2 1

NO.	Parts Number	Parts Name	Description
1	H444406A01	Rubber fitting	
2	H519533-01	Cover,cell	
3	H348112-01	Heat insulator	
4	H519530-01	Block	
5	H337618-01	Posistor	
6	H509604-01	Plate	
7	E011822900	Thermal switch	
8	U800694300	Catalyzer pipe	
9	H519531A01	Heat insulator	
11	H517345A01	Retainer,cell	

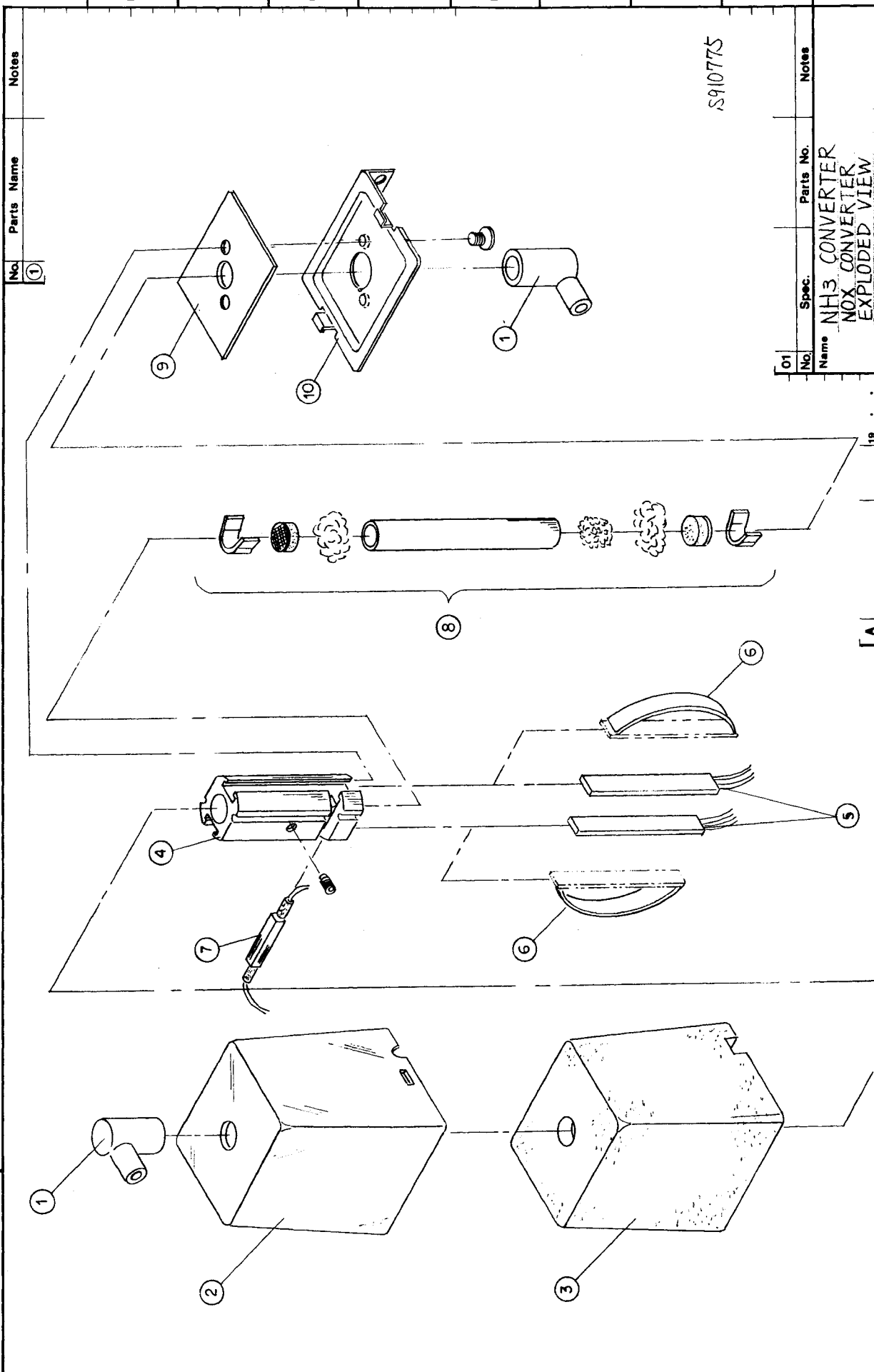


A	Drawn <i>M. Orsini</i>	Checked <i>M. Kado</i>	Revised	Approved <i>M. Kado</i>		Scale 1:1		Drawing No. S210353	Vn. sz. 	Date 19.09.2006	No. 	Spec. 	Parts No. 	Name NH3 CONVERTER CATALYZER UNIT EXPLODED VIEW.	Notes

1 2 3 4 5 6 7 8 9 10 11 12 13 14

HORIBA
Exterior Drawing

S310321
SS 6/1

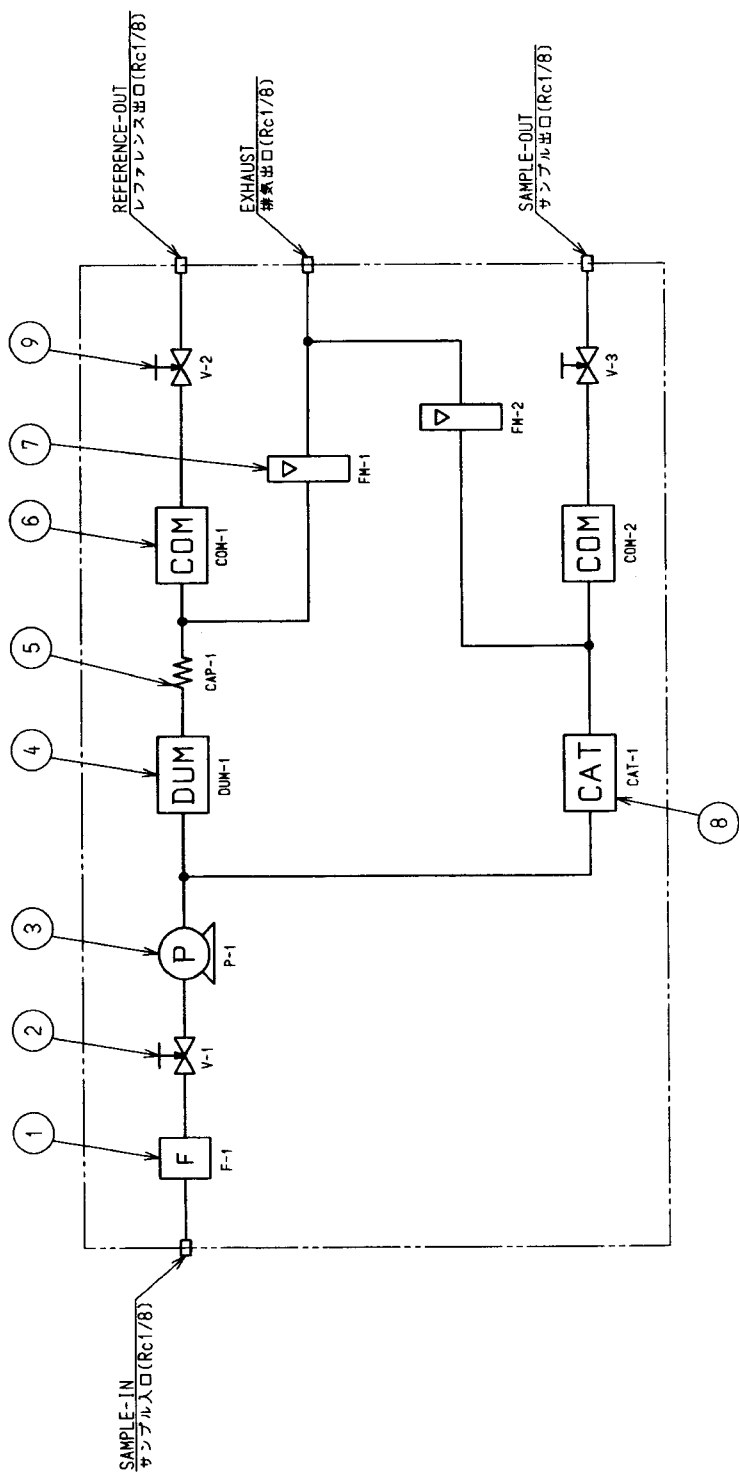


S910775

No. 1		Parts Name		Notes	
Name NH3 CONVERTER		Spec.		Parts No.	
NOX CONVERTER		Drawing No.		V3 S3	
EXPLODED VIEW		Scale		S310321	
Date 19 90 .SEP. 6		Approved		M. Onoda N. Kanda T. Yamada	
Drawn		Checked		Revised	

H03 HORIBA, Ltd.

NO.	PARTS NAME	NOTES
1	FILTER フィルタ	
2	NEEDLE VALVE ニードルバルブ	
3	PUMP ポンプ	
4	DUMMY ダミー	
5	CAPILLARY キャピラリ	
6	CONVERTER コンバータ	
7	FLOW METER 流量計	
8	CATALYZER 酸化器	
9	NEEDLE VALVE ニードルバルブ	



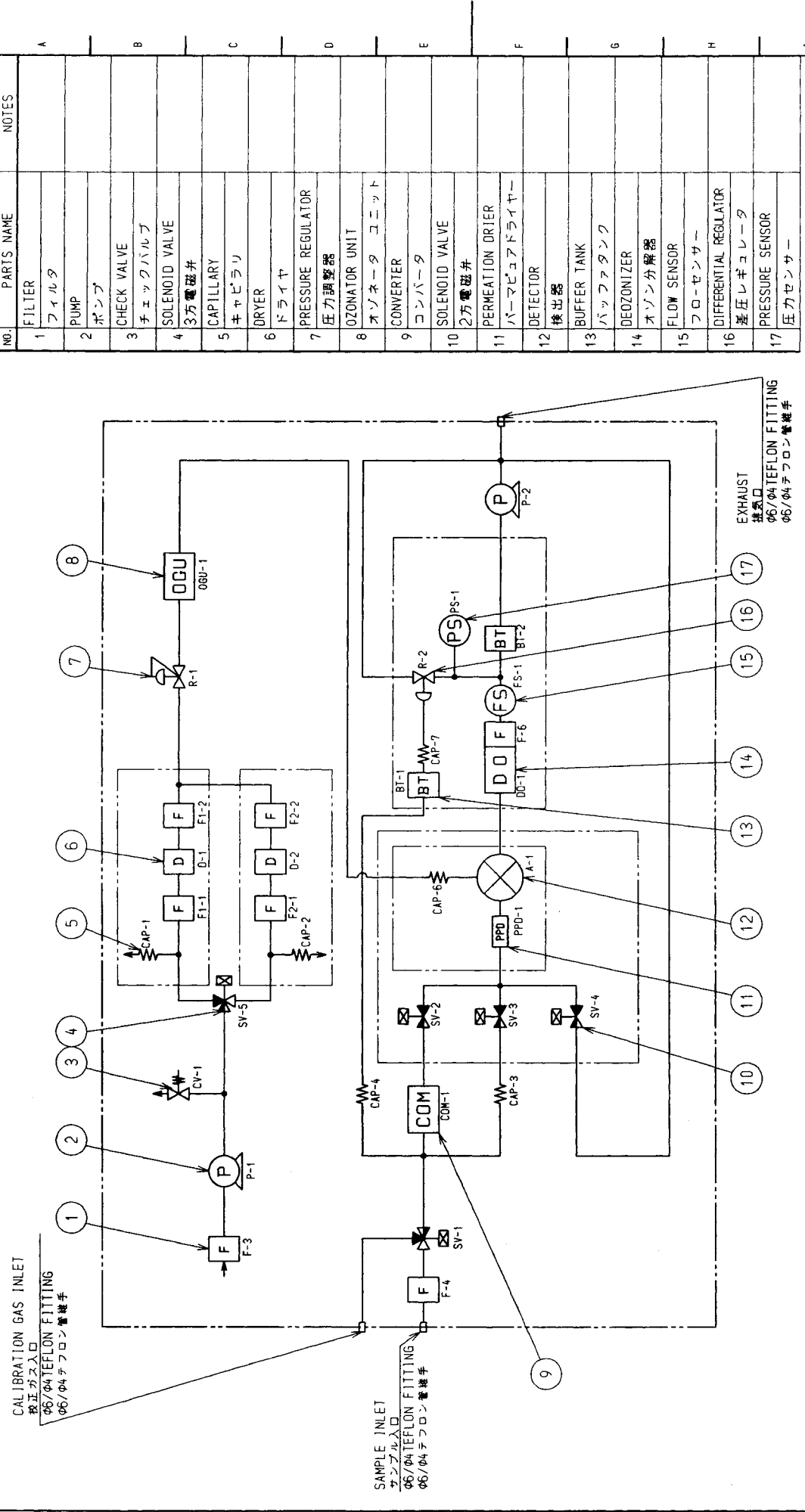
No.	SPEC.	PARTS NO.	NOTES
NAME CU-2			
FLOW SCHEMATIC OF NH3 CONVERTER UNIT 大気汚染監視用NH3コンバーターフローシート			
SCALE		FREE	
DATE		1990. JUN. 6	
DRAWN		CHECKED	APPROVED
REVISED			
S. MAKAMURA 1993.OCT.7		N. SHIMIZU 1992.FEB.6	
ニードルバルブ追加		フロー変更	
A-CAD DRAWING NO. V344816B			

B712653A

ON 9H1W8Q-A

HORIBA

EXTERIOR DRAWING



NO.	PARTS NAME	NOTES
1	FILTER	
2	フィルタ	
3	PUMP	
4	ポンプ	
5	CHECK VALVE	
6	チェックバルブ	
7	SOLENOID VALVE	
8	3方電磁弁	
9	CAPILLARY	
10	キャピラリ	
11	DRYER	
12	ドライヤ	
13	PRESSURE REGULATOR	
14	圧力調整器	
15	OZONATOR UNIT	
16	オゾネータユニット	
17	CONVERTER	
18	コンバータ	
19	SOLENOID VALVE	
20	2方電磁弁	
21	PERMEATION ORIER	
22	パーミアエドライヤー	
23	DETECTOR	
24	検出器	
25	BUFFER TANK	
26	バッファタンク	
27	DEOZONIZER	
28	オゾン分解器	
29	FLOW SENSOR	
30	フローセンサー	
31	DIFFERENTIAL REGULATOR	
32	差圧レギュレータ	
33	PRESSURE SENSOR	
34	圧力センサー	

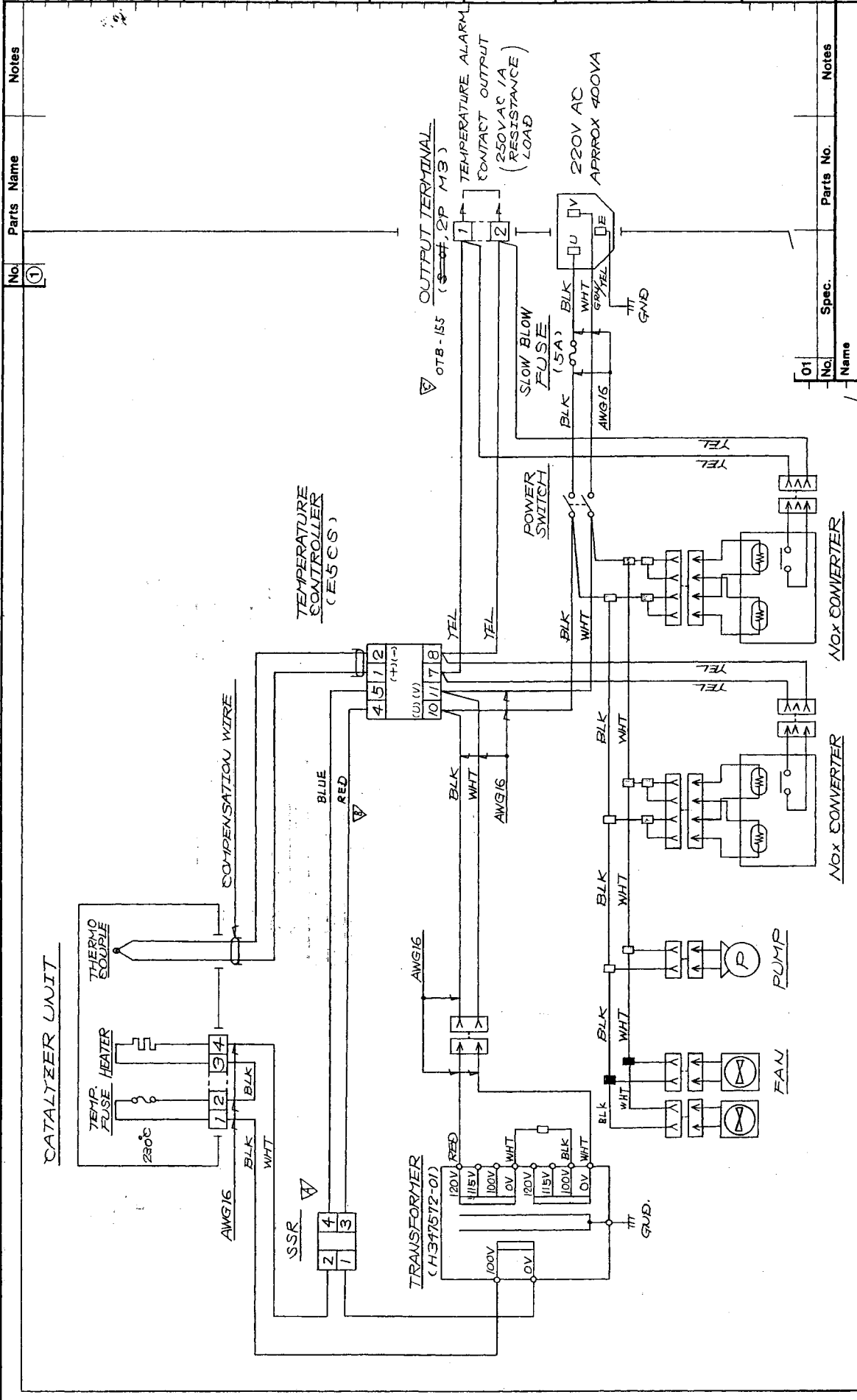
NO.	SPEC.	PARTS NO.	NOTES
1	NAME	APNA-360	
2	CHANGE OF NAME	A.KAGAWA 1994.OCT.31	
3	CHANGE OF FLOW	A.KAGAWA 1994.AUG.17	
4	SCALE	FREE	
5	DATE	1994.OCT.31	

CHANGE OF NAME	A.KAGAWA	1994.OCT.31
CHANGE OF FLOW	A.KAGAWA	1994.AUG.17
SCALE	FREE	
DATE	1994.OCT.31	
DRAWN	Y. Kogawa	
CHECKED	T. Ohnishi	
APPROVED	T. Yamada	

HOS HORIBA Ltd.

V344921

35



Name		Spec.		Parts No.		Notes	
01							
ADDED WIRE COLOR (1994 JUN 10) CHANGED WIRING (1994 JUN 29)							
Drawn		Checked		Revised		Approved	
O. Takiguchi		N. Kadoya		J. Yamada			
Date		Scale		Drawing No.		V3.53	
19 90 JUN 20		Free		V344921			

NOTE: ALL WIRE TO BE AWG 18 UNLESS OTHERWISE SPECIFIED.

HORIBA, Ltd.

2 Miyanohigashi, Kisshoin Minami-ku, Kyoto 610-8510 Japan
<http://www.horiba.com>

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